NLP Project Report

AI-Generated vs Human-Written Text Authenticity Detection

1. Problem Statement and Dataset Description

The goal of this project is to build a system that can automatically determine whether a piece of text is written by a human or generated by an AI model. This type of detection is becoming important for reducing misinformation, supporting academic integrity, and helping digital platforms validate the authenticity of content.

The dataset used in this project is a public Kaggle dataset containing 1,464 samples. Each text is labeled as either:
- 0 = Human-written
- 1 = AI-generated

The dataset is highly imbalanced, with most samples being human-written.

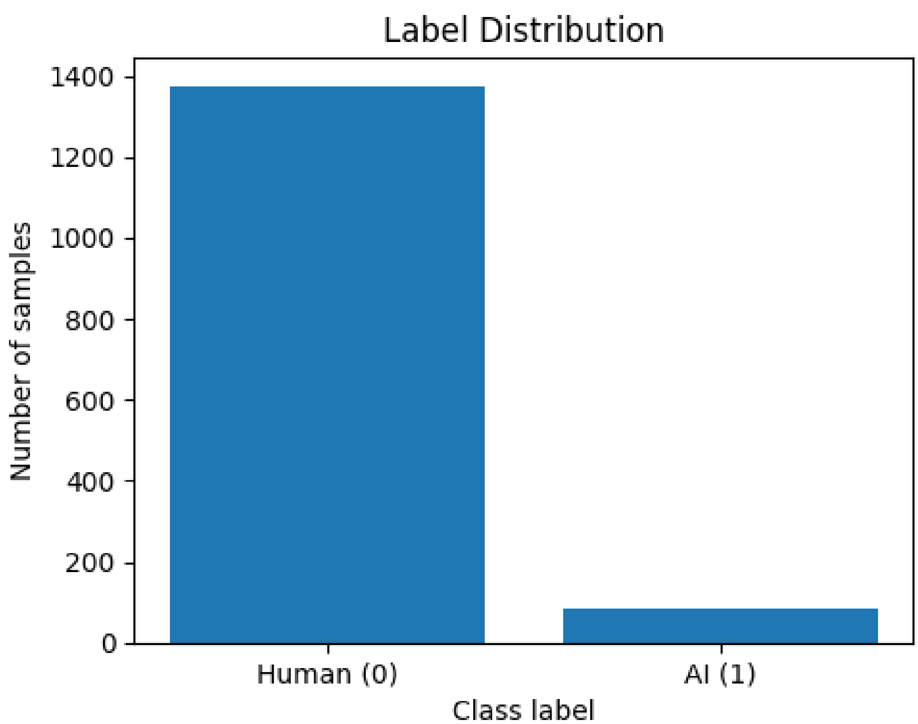

Figure 1: Label Distribution

2. Explanation of the Used Approaches

Before training the models, the text was cleaned using several preprocessing steps: lowercasing, removal of punctuation and digits, URL removal, tokenization, stopword removal, and lemmatization. This produced cleaner and more consistent input for the classifier.

The Bag-of-Words representation (CountVectorizer) was used to convert the cleaned text into numerical features. Two machine learning models were trained and compared:
1. Multinomial Naive Bayes – commonly used for text classification with count-based features.
2. Logistic Regression – a strong linear classifier that performs well on high-dimensional sparse data.

3. Performance on Unseen Data

Both models were evaluated on a held-out test set containing 292 unseen samples. Each model achieved an accuracy of 0.9932. This shows that Bag-of-Words features were effective enough for distinguishing between human and AI-generated texts in this dataset.

The confusion matrices for both models show identical behavior:
- All 275 human-written samples were correctly classified.
- 15 out of 17 AI-generated samples were correctly classified.

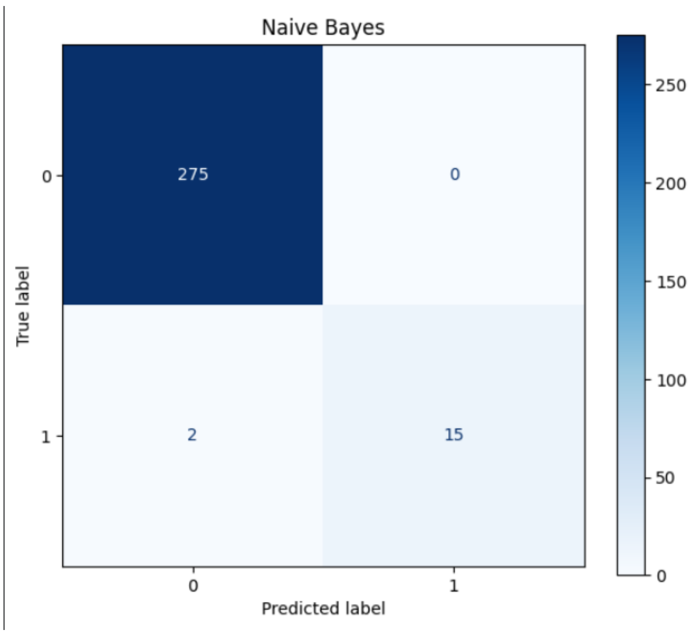

Figure 2: Naive Bayes Confusion Matrix

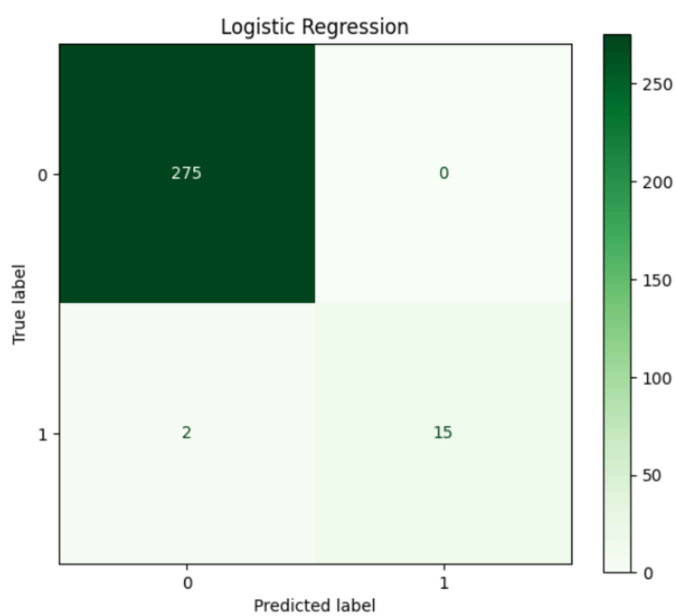

Figure 3: Logistic Regression Confusion Matrix

4. Discussion of Results

The results show that both Naive Bayes and Logistic Regression perform almost identically on this dataset. The high accuracy is strongly influenced by the dominance of human-written samples. However, both models still performed well on the minority class, correctly identifying most AI-generated texts.

Overall, the classifiers generalize well to unseen data, and the preprocessing pipeline played a key role in improving text consistency and reducing noise.